

# Customer Churn Analysis

Exploratory analysis of customer behavior, engagement, and churn risk

Prepared for Customer Success, Marketing, and Product Leadership

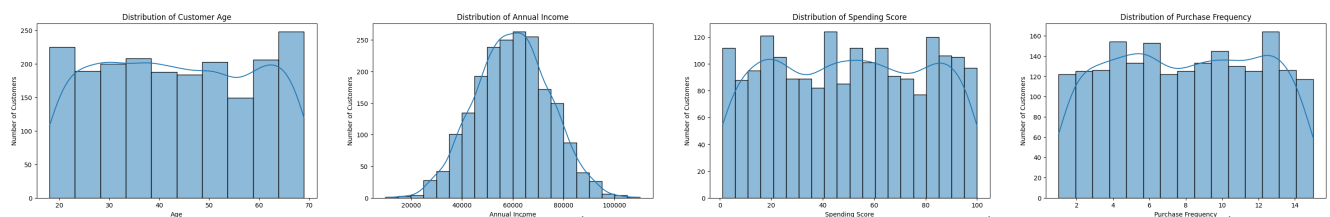
## The Churn Problem

**Roughly one-third of the customer base has already churned.** Of 2,000 customers, 665 (33.25%) have left, while 1,335 (66.75%) remain active. This report asks a single question: **what separates the customers who stay from the customers who leave?**

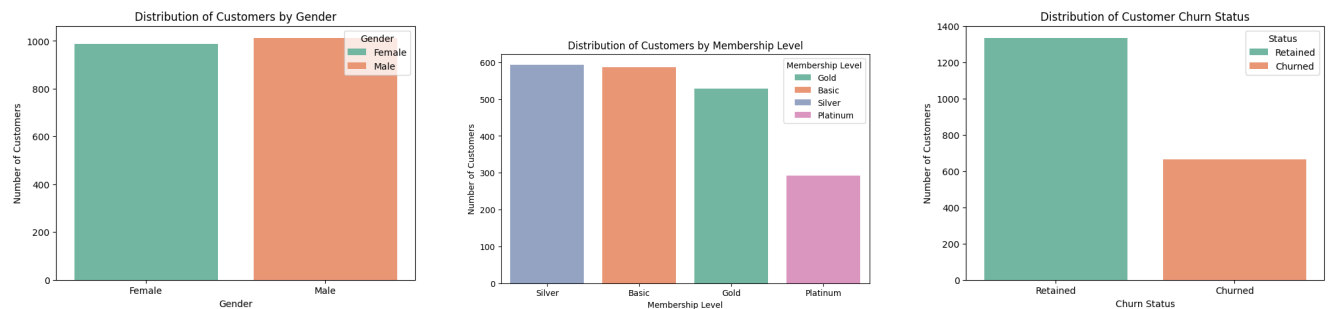
The headline finding is that the answer has very little to do with who customers are, and a great deal to do with how they behave. Behavioral engagement – spending activity, purchase frequency, and an engineered overall engagement measure – separates churned from retained customers far more clearly than age, gender, income, or membership tier. At the extreme, customers who are both **low-spending and infrequent purchasers churn at 98.46%**, while customers who are **high-spending and frequent purchasers churn at only 4.30%**.

## The Customer Base

Before turning to churn, it helps to see who the customers are. The base is broad and diverse across demographics, spending, and purchasing activity, with no single group dominating.



**Figure 1:** Customers span a broad adult age range (18–69), a middle-centered income distribution (median approx. \$60,000), and wide variation in both spending score and purchase frequency. No single band dominates any of the four distributions.



**Figure 2:** The customer base is almost evenly split by gender (1,012 male, 988 female) and spread across four membership tiers, with Silver the largest (593) and Platinum the smallest (292). The rightmost panel shows the churn baseline this report investigates: 33.25% of customers have churned.

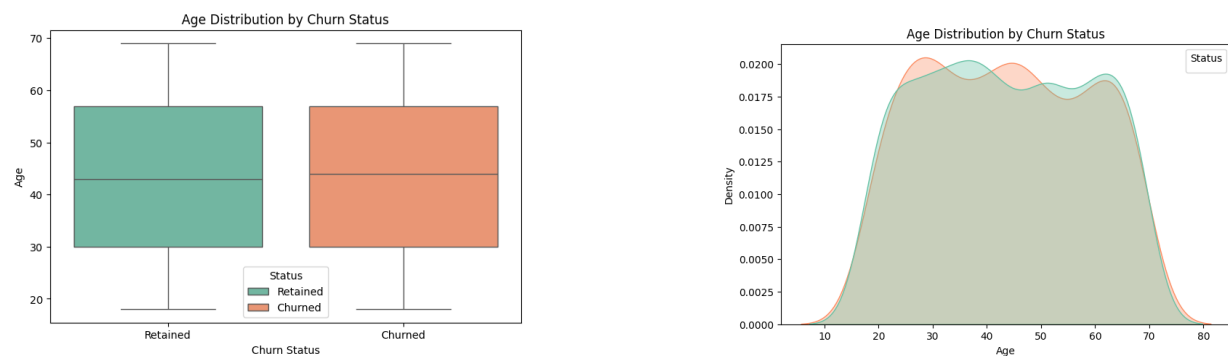
The customer base is diverse, but as the next section shows, **the churn story is not primarily demographic.**

## What Separates Customers Who Stay From Customers Who Leave?

A natural first instinct is to look for differences in demographics or income. The data does not support that instinct.

**The strongest differences emerge in customer behavior** – what customers actually buy and how often.

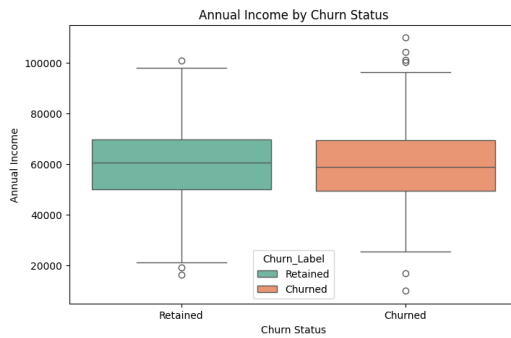
### Age and income: little separation



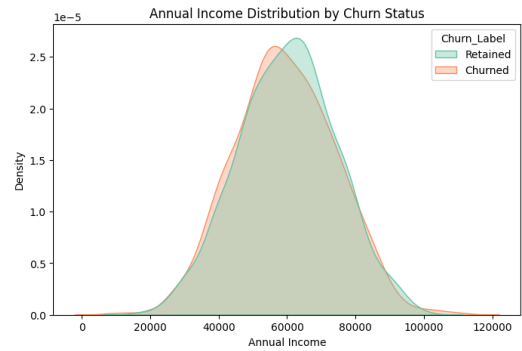
Age by churn status (box plot).

Age by churn status (density).

**Figure 3:** Retained and churned customers have nearly identical age profiles. Both groups have a median age in the early 40s, and the two density curves overlap almost completely. Age does not meaningfully distinguish who churns.



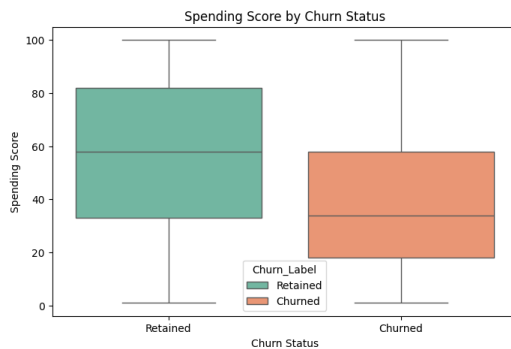
Annual income by churn status (box plot).



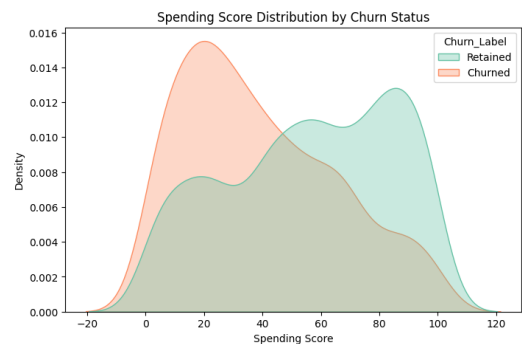
Annual income by churn status (density).

**Figure 4:** Retained and churned customers also show highly similar income distributions, both centered near \$60,000 with substantial overlap. As with age, income alone provides little separation between the two groups.

### Spending and purchase frequency: strong separation

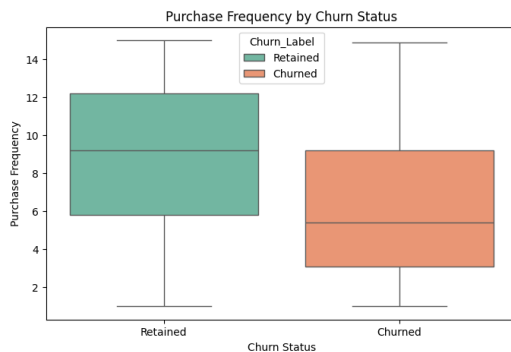


Spending Score by churn status (box plot).

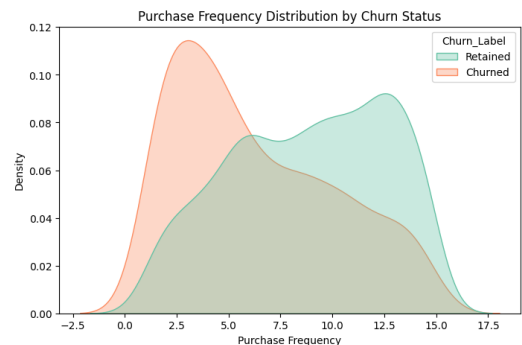


Spending Score by churn status (density).

**Figure 5:** Here the pattern changes sharply. Retained customers have a substantially higher median Spending Score (approximately 60) than churned customers (approximately 34), and the density plot shows churned customers concentrated at markedly lower spending levels.



Purchase Frequency by churn status (box plot).



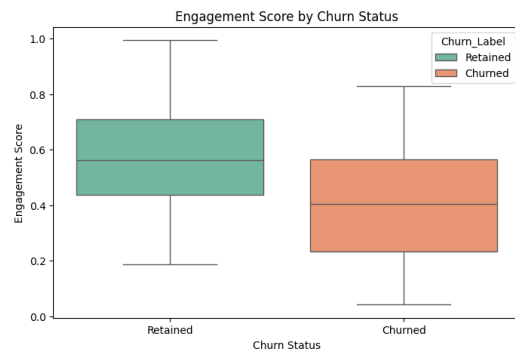
Purchase Frequency by churn status (density).

**Figure 6:** Churned customers also purchase noticeably less often: retained customers have a higher median purchase frequency (approximately 9) compared with churned customers (approximately 5). The density plot shows churned customers shifted clearly toward low-frequency purchasing.

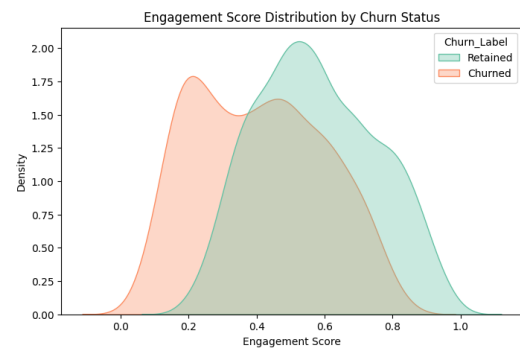
The visual progression across this page is deliberate: age and income show **little separation**, while spending and purchase frequency show **strong separation**. **Behavioral engagement, rather than demographics or income, provides the clearest separation between churned and retained customers.**

## Where Churn Risk Concentrates

This is the central page of the analysis. An engineered **Engagement Score** – combining spending and purchase frequency into a single measure – produces the widest and most useful churn split found anywhere in the data.



Engagement Score by churn status (box plot).

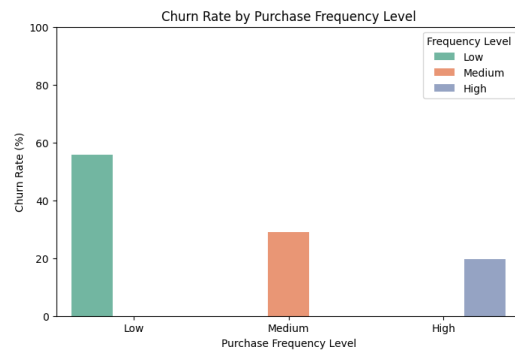


Engagement Score by churn status (density).

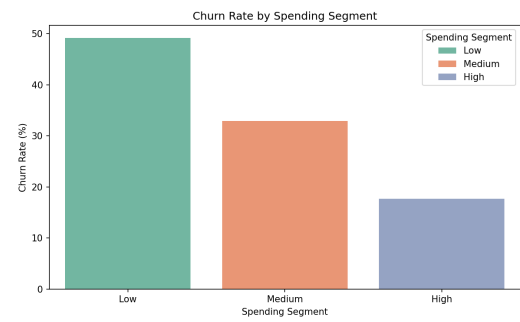
**Figure 7:** Retained customers have a substantially higher median Engagement Score (approximately 0.55) than churned customers (approximately 0.40). When segmented into tiers, **Low-engagement customers churn at 71.62%, versus only 14.26% for High-engagement customers** — a 57.4 percentage-point gap, the broadest single segmentation signal in the analysis.

## The behavioral risk ladder

The same stepped pattern appears independently in purchase frequency and spending, reinforcing that this is a broad behavioral phenomenon rather than an artifact of one measure.



Churn rate by Purchase Frequency Level.



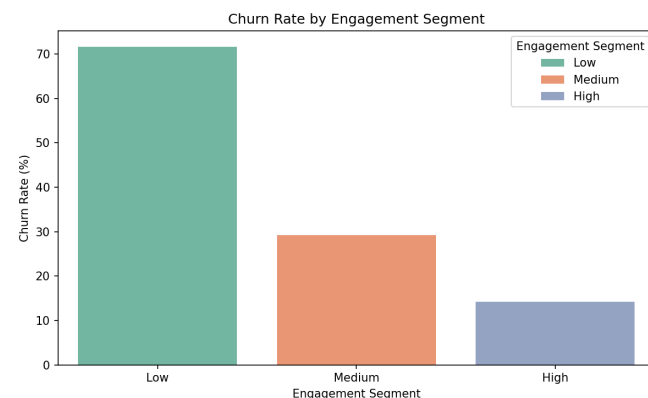
Churn rate by Spending Segment.

**Figure 8:** Churn falls consistently as purchase frequency rises: **55.65% (Low) → 29.09% (Medium) → 19.66% (High)**. Spending shows the identical stepped pattern: **49.18% (Low) → 32.93% (Medium) → 17.66% (High)**. Both behavioral gradients point toward the same conclusion: declining activity is an early-warning signal.

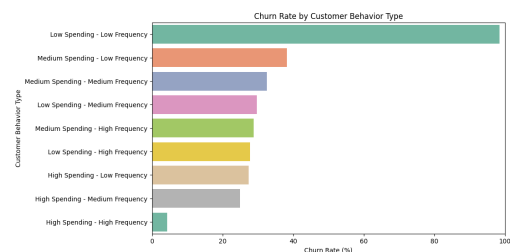
## The clearest high-risk and low-risk segments

Combining spending and frequency into a single **Customer Behavior Type** produces the sharpest contrast in the entire dataset:

|                                |              |
|--------------------------------|--------------|
| Low Spending – Low Frequency   | 98.46% churn |
| High Spending – High Frequency | 4.30% churn  |



Churn rate by Engagement Segment.



Churn rate by Customer Behavior Type.

**Figure 9:** Engagement Segment (left) mirrors the same Low → High risk gradient seen throughout this page. Customer Behavior Type (right) shows the single most extreme finding in the analysis: customers who are both low-spending and infrequent purchasers churn at a near-certain **98.46%**, while customers strong on both dimensions churn at just **4.30%** — a 94.2 percentage-point spread, against an overall base rate of 33.25%.

Churn risk is concentrated among customers with weak engagement, while customers with strong spending and purchasing activity show much stronger retention.

| Customer Group     | Segment                      | Customers | Churn Rate |
|--------------------|------------------------------|-----------|------------|
| Purchase Frequency | Low                          | 566       | 55.65%     |
|                    | Medium                       | 722       | 29.09%     |
|                    | High                         | 712       | 19.66%     |
| Spending           | Low                          | 667       | 49.18%     |
|                    | Medium                       | 665       | 32.93%     |
|                    | High                         | 668       | 17.66%     |
| Engagement         | Low                          | 370       | 71.62%     |
|                    | Medium                       | 1,118     | 29.25%     |
|                    | High                         | 512       | 14.26%     |
| Behavior Type      | Low Spending–Low Frequency   | 195       | 98.46%     |
|                    | High Spending–High Frequency | 256       | 4.30%      |

**Table 1:** Overall churn rate across the full base is 33.25%. Highlighted cells mark the clearest high-risk (red) and low-risk (green) results.

## Business Implications

### Ten Business Insights

#### Strongest risk signals

- **Low purchase frequency is a major risk segment.** Customers purchasing infrequently churn at 55.65%, versus 19.66% for frequent purchasers. Customers in the 1–5 purchase range warrant targeted retention attention.
- **Low spending is strongly associated with churn.** Low-spending customers churn at 49.18%, versus 17.66% for high-spending customers, suggesting value in re-engaging low-spending customers before they become inactive.
- **Overall engagement is the strongest broad segmentation signal.** Low-engagement customers churn at 71.62% versus 14.26% for high-engagement customers – a 57.36-point gap – making the Engagement Score a potentially useful early-warning indicator.
- **The highest-risk customers are both low-spending and infrequent purchasers.** This combined segment churns at 98.46%, the single clearest high-risk population identified, and could be considered a priority for reactivation efforts.

#### The retention pattern

- **Highly active customers are exceptionally stable.** High-spending, high-frequency customers churn at just 4.30%, far below the 33.25% base rate.
- **Purchase frequency shows a clear behavioral gradient:** 55.65% → 29.09% → 19.66% from Low to High frequency, useful for tiered retention monitoring.
- **Spending shows the same stepped pattern:** 49.18% → 32.93% → 17.66% from Low to High spending, a second behavioral warning signal to monitor alongside frequency.

#### Weaker segmentation factors

- **Demographics are unlikely to be the best primary retention lever.** Age distributions are highly similar between groups; female customers churn at 35.02% versus 31.52% for male customers, a gap of only about 3.5 points.
- **Income provides little practical differentiation:** 34.00% (Low) vs. 33.30% (Middle) vs. 32.40% (High) – a spread of only 1.60 points.
- **Membership tier alone does not meaningfully distinguish churn risk:** 34.13% (Basic), 31.53% (Silver), 34.22% (Gold), 33.22% (Platinum) – a spread of only 2.69 points. Membership should be considered alongside behavioral signals, not as a primary indicator.

### Data Quality Note

The dataset contains 2,000 customer records with no missing values, no duplicate rows, and unique Customer IDs throughout. Nine Annual Income observations were flagged by the IQR method (four below \$20,099, five above \$99,351). On inspection these were plausible customer records rather than data errors, so none were removed.

### Recommended Business Focus

The data suggests that retention efforts should focus primarily on **behavioral engagement** rather than demographic or membership characteristics. Priority areas include: monitoring declining purchase frequency, monitoring low spending activity, using overall engagement as an early-warning signal, and prioritizing outreach to low-spending/low-frequency customers. Demographics, income, and membership tier appear less useful as a primary basis for churn targeting, though they may still play a supporting role alongside behavioral indicators.

### Handoff to the Data Science Team

Behavioral and engineered engagement features – particularly Engagement Score, Purchase Frequency Level, Spending Segment, and Customer Behavior Type – appear the most promising candidates for a subsequent churn-prediction model, given the clear separation shown between retained and churned customers. Income segmentation and the spending-frequency gap produced considerably weaker separation and are not recommended as primary features. Further statistical testing and predictive modeling are required to establish independent predictive value.

Source: Customer Churn dataset (2,000 records, 9 original features). All figures and statistics are drawn directly from the completed exploratory data analysis notebook.